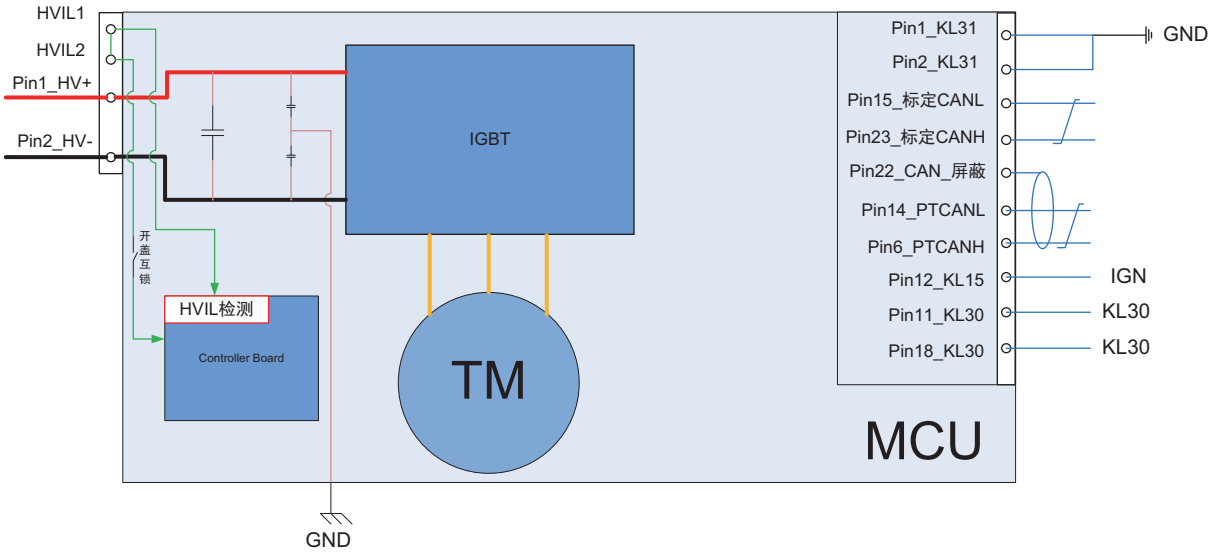
P0A40-1C、 P1A1D-24、 P1A1D-23、 P1A4F-00

Fault diagnosis code
P0A40-1C：Resolver signal out of range
P1A1D-24：Resolver signal amplitude exceeds the upper limit
P1A1D-23：Resolver signal amplitude exceeds lower limit
P1A4F-00：Resolver signal zero drift overrun
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Motor resolver fault • Resolver signal circuit of PEU fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0A40-1C
The absolute value of resolver sin/cos signal exceeds the threshold （140%）
P1A1D-24
The absolute value of resolver sin/cos signal exceeds the ceiling （130%）
P1A1D-23
The absolute value of resolver sin/cos signal exceeds the lower limit （60%）
P1A4F-00
The offset deviation of resolver sin/cos signal exceeds the limit value （10%）
To set the effect of a fault code condition
Reduction of power severity

Circuit diagram



EV901001

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of PEU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the PEU with diagnostic apparatus. 4 Is there any other fault code except for P0A40-1C、P1A1D-24、P1A1D-23、P1A4F-00?
		YES → Refer to:DTC Summary list(PEU).
NO ↓		
Step 3	Check the connecting cable of PEU.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of PEU. 5 Check whether the PEU connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check resolver signal circuit
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check whether the resolver signal circuit exist the situation of open or short circuit. 5 Check whether the resolver signal circuit exist the situation of shorting to battery. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check resolver resistance.
	1 Measure resolver sin resistance. 2 Measure resolver cos resistance. 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Check resolver signal checking module.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check whether the resolver signal checking module is damaged. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Test whether PEU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1002 of PEU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
	YES → Change the PEU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	